UBT

Computer Science and Engineering

Machine Learning Models Lab Report

Diabetes Prediction using Classification and Clustering Techniques

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1. Software Environment

Programming Language: Python 3.x

Libraries and Versions:

- numpy: For numerical computations and array operations
- pandas: For data manipulation and analysis
- matplotlib: For data visualization and plotting
- eaborn: For statistical data visualization
- cikit-learn: For machine learning algorithms and evaluation metrics
- train_test_split: Data splitting functionality
- StandardScaler: Feature scaling
- MLPClassifier: Neural Network implementation
- Perceptron: Perceptron algorithm
- KNeighborsClassifier: k-Nearest Neighbors algorithm
- DecisionTreeClassifier: Decision Tree algorithm
- KMeans: K-means clustering algorithm
- SelectKBest, f_classif, RFE: Feature selection methods
- RandomForestClassifier: For feature importance analysis
- LeaveOneOut: For cross-validation
- Various metrics: classification_report, confusion_matrix, accuracy_score, adjusted_rand_score
- tqdm: For progress bars during cross-validation

2. Dataset and Preprocessing

Dataset Description

- Dataset: Pima Indians Diabetes Dataset
- Samples: 768 patients
- Features: 8 clinical features
    - pregnancies: Number of pregnancies
    - glucose: Plasma glucose concentration
    - blood pressure: Diastolic blood pressure
    - skin thickness: Triceps skin fold thickness
    - insulin: 2-Hour serum insulin
    - bmi: Body mass index
    - diabetes pedigree: Diabetes pedigree function
    - age: Age in years

Target: Binary classification (0: Non-diabetic, 1: Diabetic)

Data Preprocessing

- Missing Value Handling:Invalid zero values in glucose, blood pressure, skin thickness, insulin, and bmi were replaced with NaN and then filled with median values
- Feature Scaling: StandardScaler applied to normalize all features (mean=0, std=1)
- Train-Test Split:70% training, 30% testing (random state=42)

3. Classification Analysis

3.1 Implemented Classifiers

Neural Network (MLP Classifier)

- Architecture: Multi-layer perceptron with hidden layers (32, 16, 8)
- Activation Function: ReLU (Rectified Linear Unit)
- Solver: Adam optimizer
- Max Iterations (maxEpochs): 600
- Early Stopping: True

Topology Explanation:

- Input layer: 8 neurons (one for each feature)
- First hidden layer: 32 neurons
- Second hidden layer: 16 neurons
- Third hidden layer: 8 neurons
- Output layer: 2 neurons (binary classification)
- Progressive reduction in layer size allows for hierarchical feature learning
- Early stopping to stop training the model when no improvement is made

Perceptron

- Max Iterations (maxEpochs):1000
- Algorithm: Single-layer linear classifier
- Random State: 42 for reproducibility

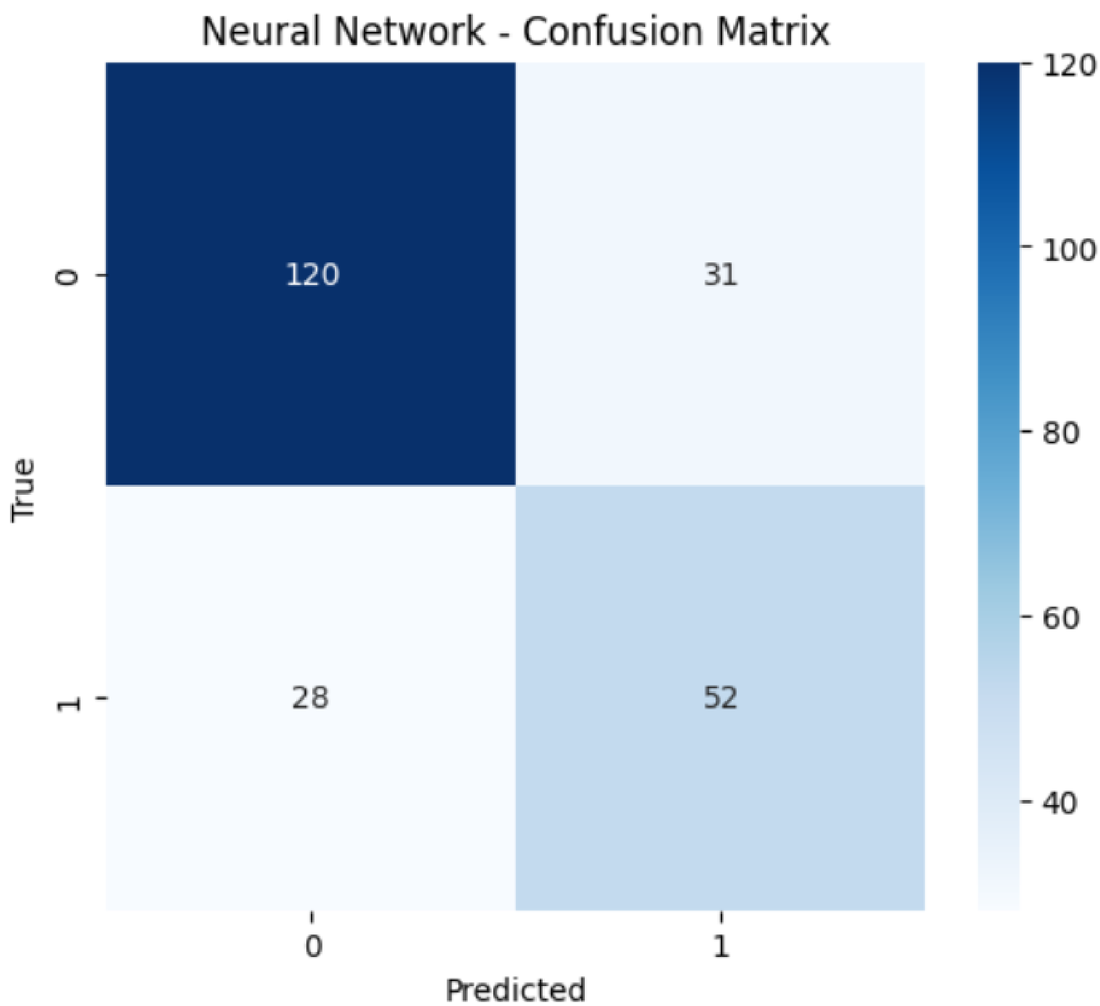
k-Nearest Neighbors (k-NN)

- Number of Neighbors (k): 5
- Distance Metric: Euclidean distance (default)
- Algorithm: Brute force neighbor search

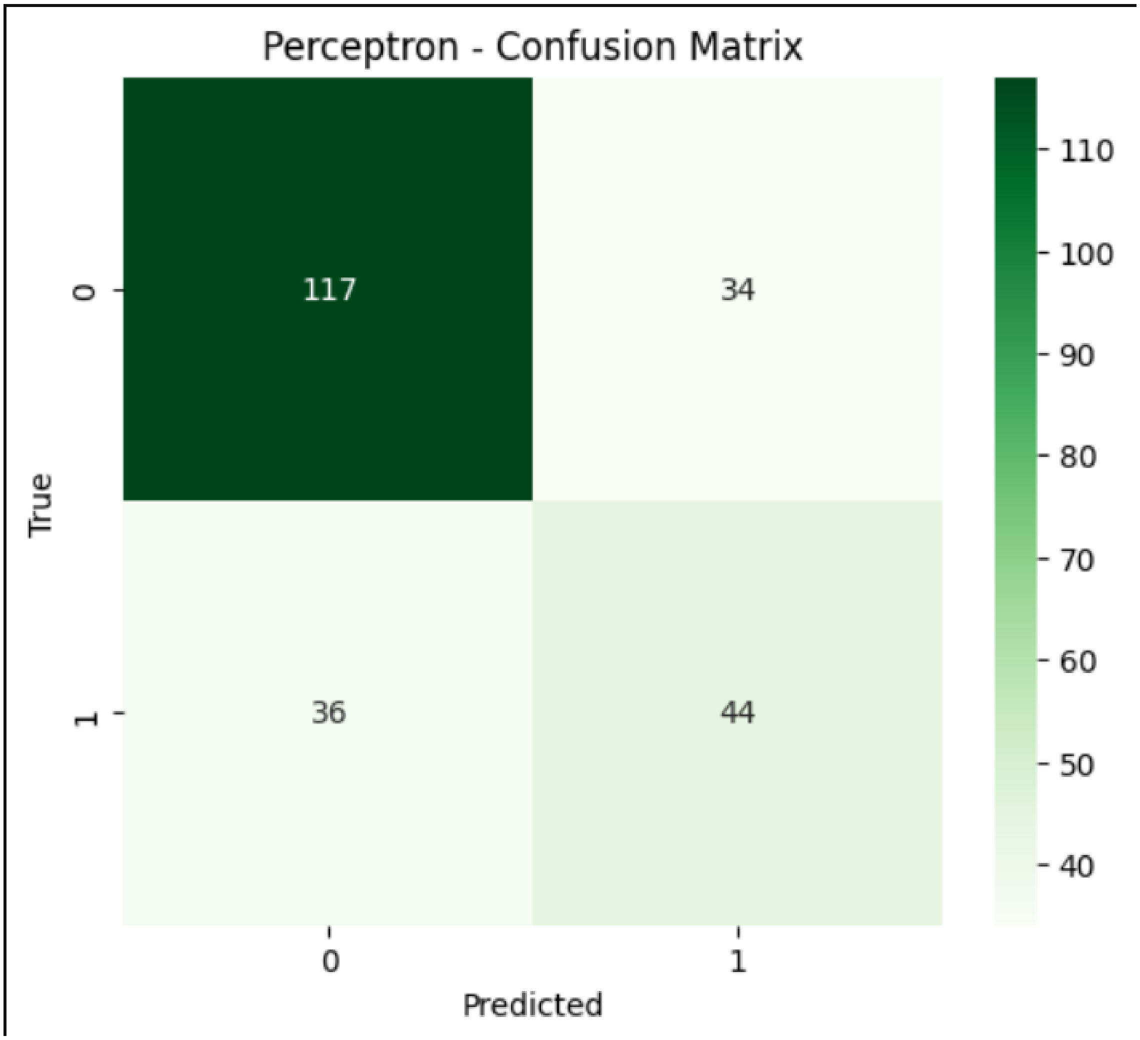
Decision Tree

- -Splitting Criterion: Gini impurity (default)
- -Random State: 42 for reproducibility
- -No maximum depth specified (grows until pure leaves)

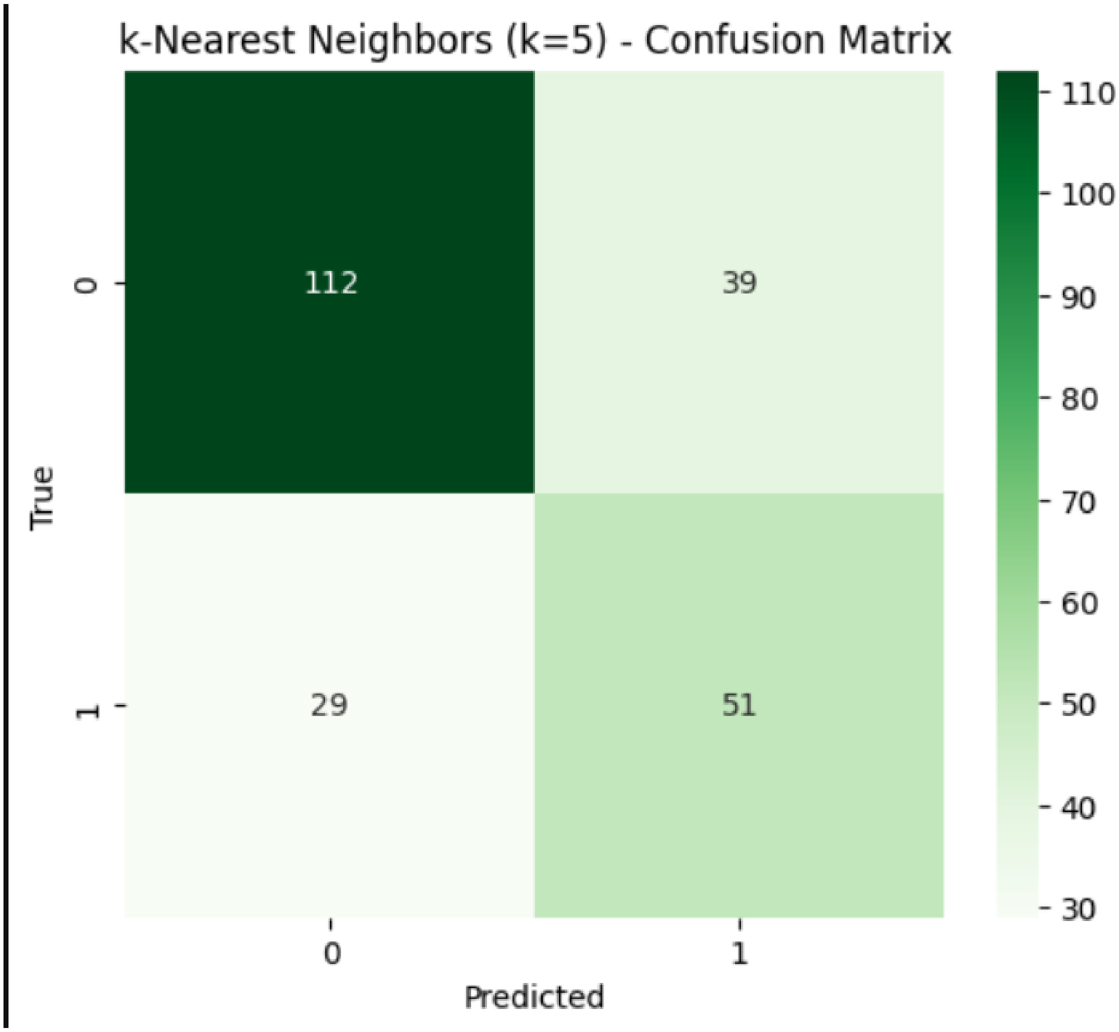
3.2 Classification Results



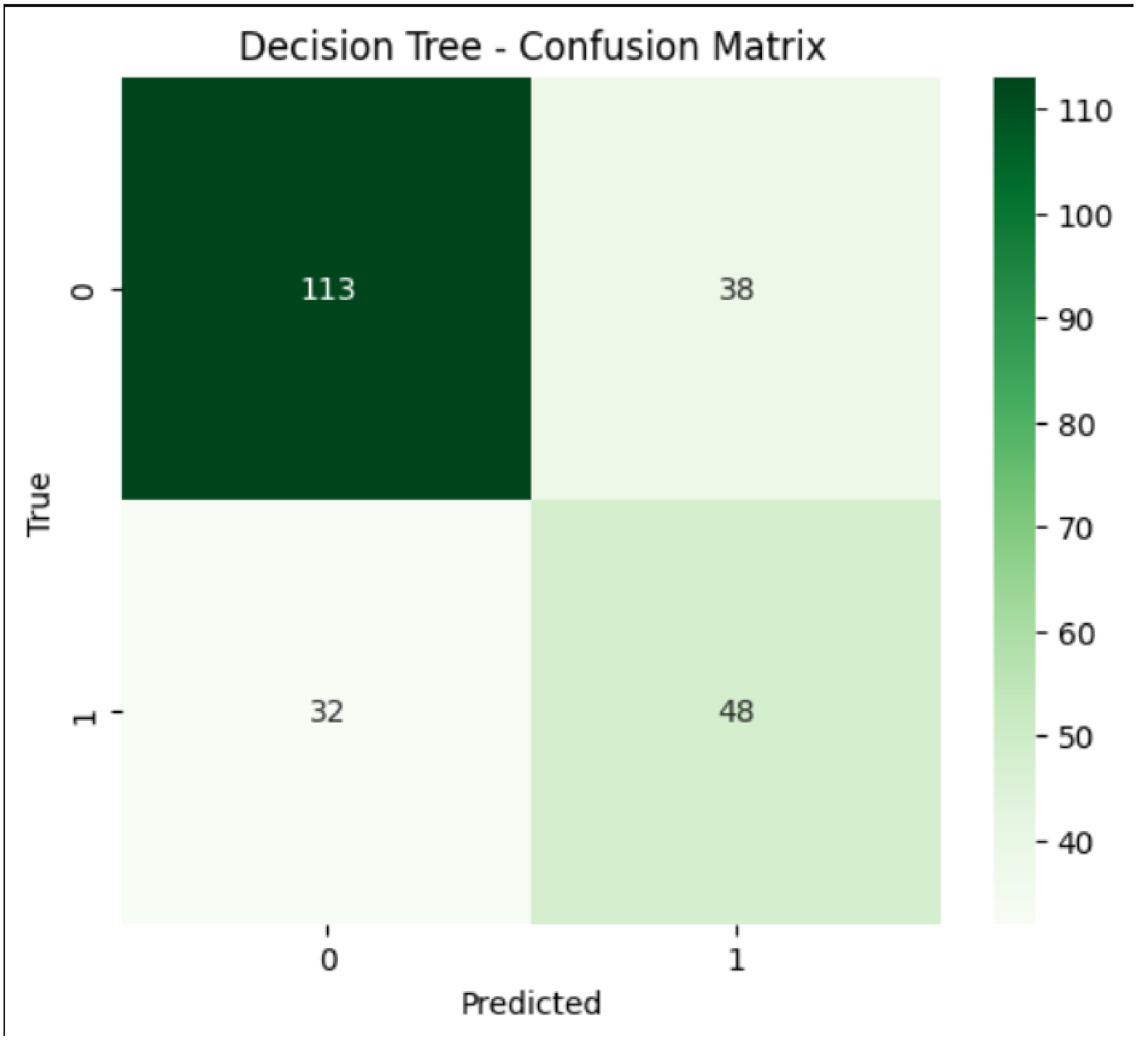
Neural Network Confusion Matrix



Perceptron Confusion Matrix



k-NN Confusion Matrix



Decision Tree Confusion Matrix

Performance Summary Table

Model	Accuracy	Precision	Recall	F1-Score
Neural Network	0.7446	0.6265	0.6500	0.6380
Perceptron	0.6970	0.5641	0.5500	0.5570
k-NN	0.7056	0.5667	0.6375	0.6000
Decision Trees	0.6970	0.5581	0.6000	0.5783

Best Performing Model: Neural Network achieved the highest accuracy (74.46%) and F1-score (63.80%)

3.3 Leave-One-Out Cross-Validation Results

LOOCV was performed on the training set to assess model stability:

Model	LOOCV Accuracy
Perceptron	0.7095
k-NN	0.7542
Decision Trees	0.7132

Analysis: k-NN showed the most stable performance under LOOCV, suggesting good generalization capability.

4. Feature Selection Analysis

4.1 Feature Selection Methods Implemented

Method 1: ANOVA F-test (SelectKBest)

- Approach: Univariate statistical test
- Metric: F-statistic measuring relationship between each feature and target
- Configurations: Top 4, 6, and 8 features selected

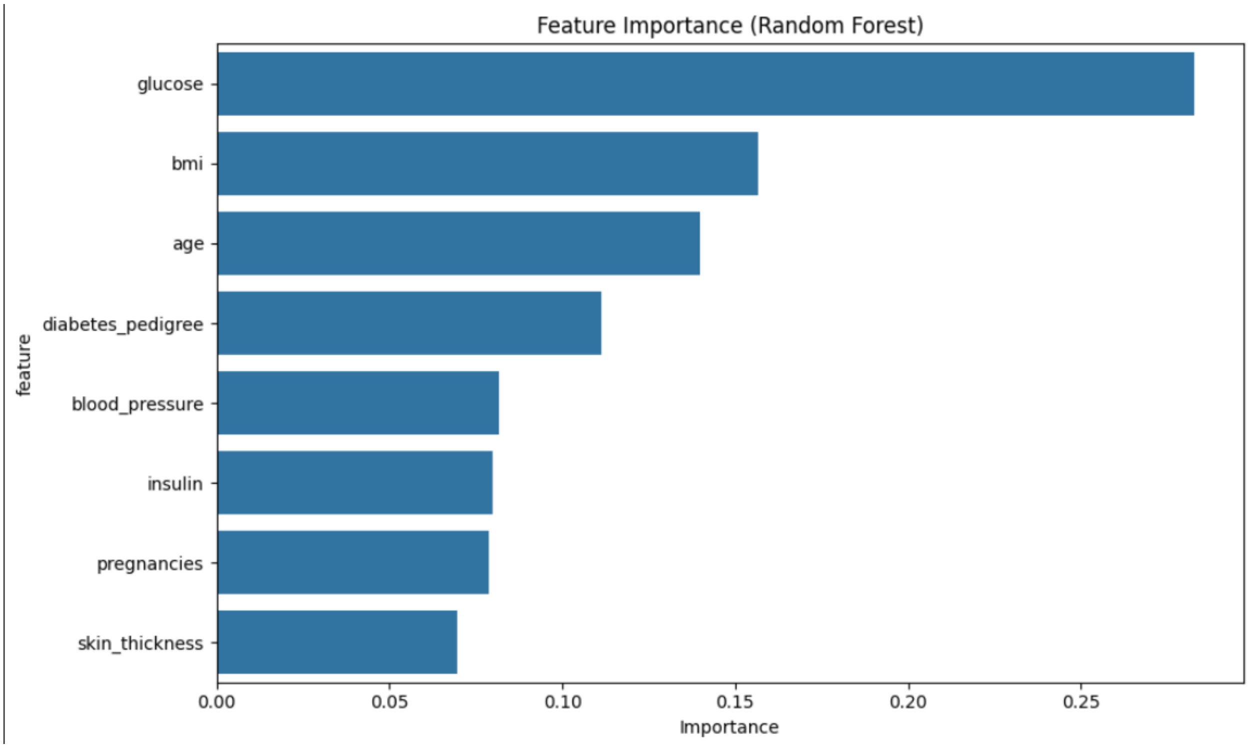
Method 2: Recursive Feature Elimination (RFE)

- Approach: Iterative feature elimination using Random Forest as estimator
- Process: Recursively removes least important features
- Configurations: 4 and 6 features selected

Method 3: Feature Importance Analysis

- Method: Random Forest feature importance scores
- Metric: Mean decrease in impurity across all trees

4.2 Feature Selection Results



Feature Importance Bar Chart

Feature Importance Ranking:

- glucose: 0.2829 (highest importance)
- bmi: 0.1565
- age: 0.1397
- diabetes pedigree: 0.1113
- blood pressure: 0.0817
- insulin: 0.0797
- pregnancies: 0.0786
- skin thickness: 0.0696 (lowest importance)

Impact of Feature Selection on Performance:

Method	Features Selected	Accuracy
All Features	All 8 Features	0.7446
ANOVA F-test (K=4)	glucose, insulin, bmi, age	0.7576

ANOVA F-test (K=6)	pregnancies, glucose, skin_thickness, insulin, bmi, age	0.7013
ANOVA F-test (K=8)	All features	0.7446
RFE (4 features)	glucose, bmi, diabetes_pedigree, age	0.7532
RFE (6 features)	glucose, blood_pressure, insulin, bmi, diabetes_pedigree,	0.7532

4.3 Feature Selection Discussion

Does using fewer features improve classifier performance?

Yes, using fewer features can improve performance:

- Best result: K=4 ANOVA F-test achieved 75.76% accuracy (improvement of 1.3%)
- Optimal feature count: 4 features appear to be optimal for this dataset
- Selected features: glucose, insulin, bmi, age consistently appear in top-performing subsets

Rigorous approach for choosing best features:

1. Statistical significance: ANOVA F-test identifies features with strongest statistical relationship to target

2. Recursive elimination: RFE removes redundant features iteratively

3. Tree-based importance: Random Forest provides feature importance based on predictive power

4. Cross-validation: Validate feature subsets using LOOCV to ensure generalization

Key findings:

- Glucose level is consistently the most important feature
- Clinical measures (BMI, age) are more predictive than demographic features
- Removing less informative features (skin thickness, pregnancies) can reduce noise and improve performance

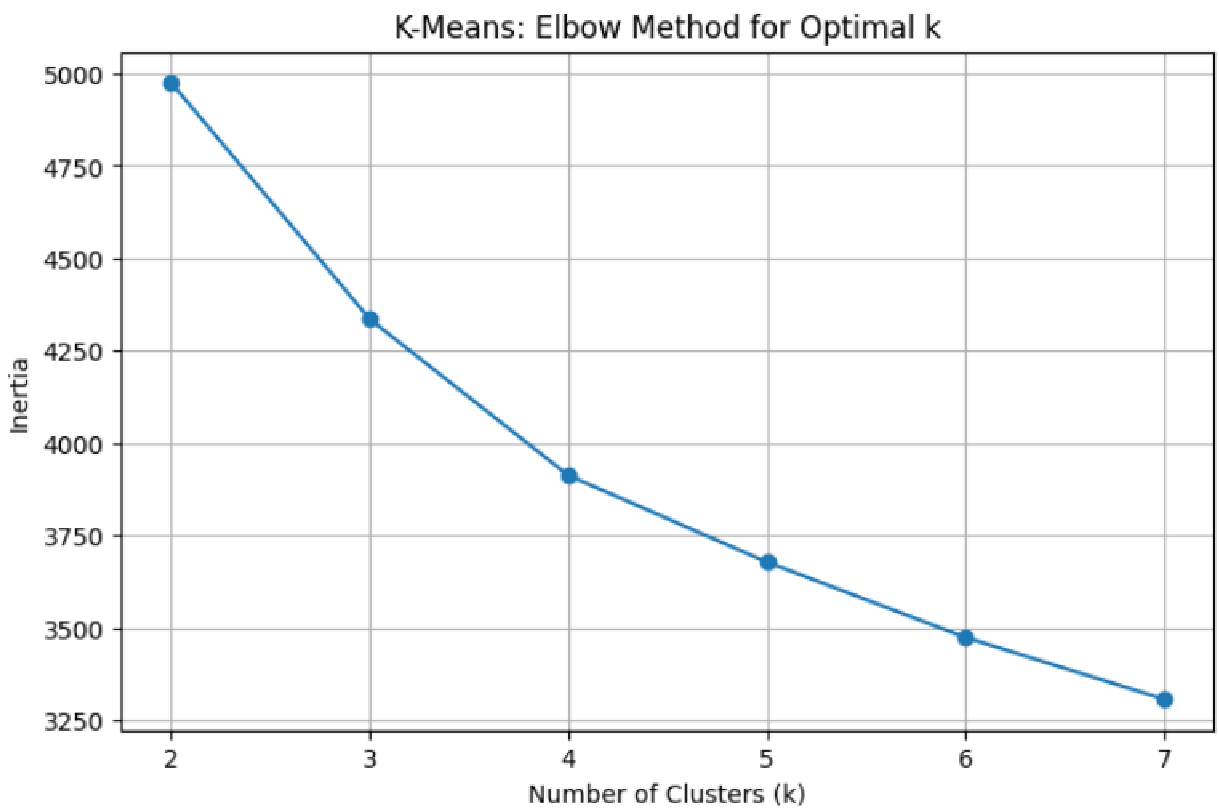
5. Clustering Analysis

5.1 K-Means Clustering Implementation

Preprocessing for Clustering:

- Class labels removed as required
- Features standardized using separate scaler
- All 8 features used for clustering

5.2 Optimal k Selection



Elbow Method Plot showing Inertia vs Number of Clusters

5.3 Final Clustering Results (k=3)

Cluster Distribution:

- Cluster 0: 216 samples (28.1%)
- Cluster 1: 337 samples (43.9%)
- Cluster 2: 215 samples (28.0%)

Cluster Characteristics:

Final K-Means Clustering (k=3)

Cluster distribution:

Cluster 0: 248 samples

Cluster 1: 335 samples

Cluster 2: 185 samples

Cluster Centers:

Cluster 0: [1.0470787   0.30028055 0.49869482 0.10381752 -0.01203346 0.05147404

-0.04537878 1.08907535]

Cluster 1: [-0.46459069 -0.49428384 -0.4651046 -0.5579768 -0.31670965 -0.57583692

-0.18276281 -0.62093654]

Cluster 2: [-0.56236561 0.49251627 0.17369582 0.87121882 0.58963261 0.97372869

0.39178096 -0.33555105]

Inertia: 4335.74

5.4 Clustering vs True Labels Comparison

Cluster Composition Analysis:

Cluster	**Total Samples**	**Non-diabetic**	**Diabetic**	**Diabetic %**
0	216	106 (49.1%)	110(50.9%)	50.9%
1	337	287(85.2%)	50(14.8%)	14.8%
2	215	107(49.8%)	108(50.2%)	50.2%

Adjusted Rand Index (k=3): 0.1219

5.5 Clustering Discussion

Findings:

- Cluster 1 appears to represent the "healthy" group with 85.2% non-diabetic patients
- Clusters 0 and 2 show mixed populations with approximately equal diabetic/non-diabetic distributions

- Low ARI (0.1219) indicates clustering doesn't perfectly align with true diabetes labels

Interpretation:

- K-means clustering identifies natural groupings in the data that partially correspond to diabetes status
- The algorithm discovers patterns beyond simple diabetic/non-diabetic classification
- Clinical significance: Clusters may represent different risk profiles or disease subtypes

6. Critical Discussion

6.1 Model Performance Analysis

Best Performing Classifier: Neural Network

- Achieved highest accuracy (74.46%) and F1-score (63.80%)
- Benefits from non-linear activation functions and multiple hidden layers
- Capable of learning complex feature interactions

Comparison Insights:

- Linear models (Perceptron) showed lower performance, suggesting non-linear relationships in data
- k-NN performed well, indicating local neighborhood patterns are informative
- Decision Tree moderate performance but provides interpretable rules

6.2 Feature Selection Impact

Key Insights:

- Glucose level emerges as the strongest predictor across all methods
- Feature reduction to 4 variables improved performance, suggesting some features add noise
- Clinical features (BMI, age) more predictive than demographic features

Methodological Considerations:

- Multiple feature selection approaches provide consistent rankings
- Cross-validation essential to avoid overfitting to feature subsets

6.3 Clustering Analysis

Limitations:

- Unsupervised clustering doesn't perfectly align with supervised labels (ARI=0.12)
- K-means assumes spherical clusters, which may not match data structure
- Clinical interpretation requires domain expertise

Strengths:

- Identifies meaningful patient subgroups
- Cluster 1 represents clear low-risk population
- Could inform personalized treatment strategies

6.4 Methodological Strengths

1. Proper data preprocessing: Handling of missing values and feature scaling
2. Comprehensive evaluation: Multiple algorithms and evaluation metrics
3. Cross-validation: LOOCV provides robust performance estimates
4. Feature analysis: Systematic approach to feature selection
5. Reproducibility: Fixed random states ensure consistent results

7. Conclusions

7.1 Key Findings

1. Neural networks achieved best classification performance (74.46% accuracy)
2. Feature selection improved performance: 4 features optimal (75.76% accuracy)
3. Glucose level is the most important predictor across all analyses
4. K-means clustering identified meaningful patient subgroups with partial correspondence to diabetes status
5. LOOCV confirmed k-NN as the most stable classifier

7.2 Practical Implications

- Clinical screening: Focus on glucose, BMI, age, and insulin for diabetes prediction
- Resource allocation: 4-feature model could enable simplified screening protocols
- Patient stratification: Clustering reveals distinct risk profiles for personalized care

7.3 Technical Achievements

- Successfully implemented and compared 4 different classification algorithms
- Demonstrated systematic feature selection methodology
- Applied unsupervised learning to discover data patterns
- Provided comprehensive evaluation using multiple metrics and cross-validation

This analysis demonstrates the effectiveness of machine learning approaches for diabetes prediction and patient stratification, with practical implications for clinical decision-making.